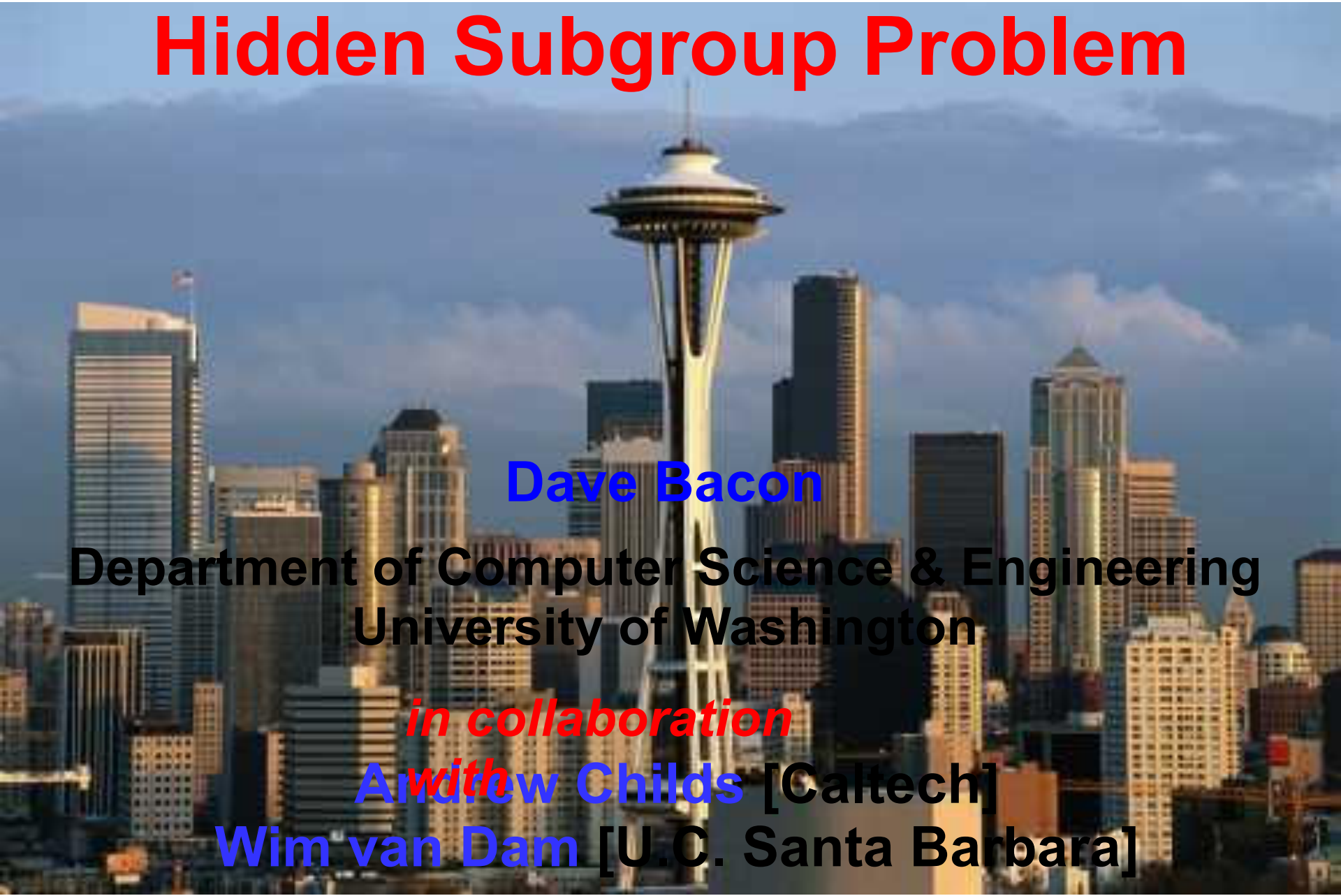
Tht Yuo Cna Reda Ths Snetnce
Impls Tat Yor Mnid Cna Efncenlty Solev
NP-Cmoplet Prbolems???

New Quantum Algorithms for the Hidden Subgroup Problem



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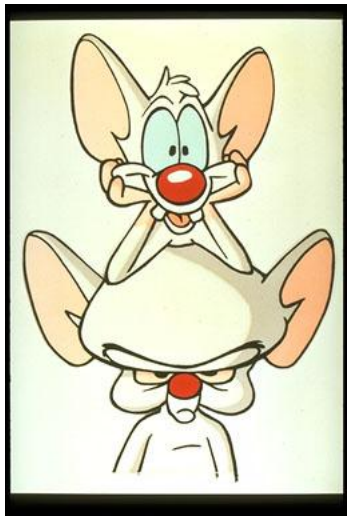
in collaboration with

Andrew Childs [Caltech]
Wim van Dam [U.C. Santa Barbara]

Quantum Algorithms

"If I had a good idea in quantum algorithms, that is what I'd be working on." – John Preskill 2004





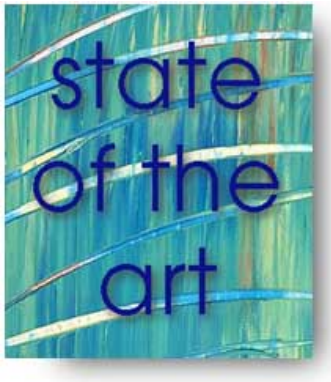
What is the future of quantum computing
if there aren't more important quantum algorithms?

When I was a Grad Student…

Quantum Algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]

State of the Art



Quantum Algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Various hidden subgroup problems [Long List]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03 Childs, van Dam 05]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …

The Problem

The Hidden Subgroup Problem (HSP)

Problem: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that is constant and distinct on left cosets of H in G. We say that f hides H.

Goal: Find (a generating set for) H.

Efficient algorithm: run time $\text{poly}(\log |G|)$.

Groups

Group Definition

Set of elements with a binary operation "$\cdot$"

Binary operation satisfies:

1. Closure: $g1 \cdot g2$ is in the set
2. Associativity: $(g1 \cdot g2) \cdot g3 = g1 \cdot (g2 \cdot g3)$
3. Identity element: Exists e such that $g1 \cdot e = g1$
4. Inverse: For all g1 there exists an element g2 such that $g1g2 = e$.

We will study only finite groups (set is finite.)

Left Cosets

Subgroups

Subset of group which is itself a group.

Left Cosets

Let H be a subgroup of G.

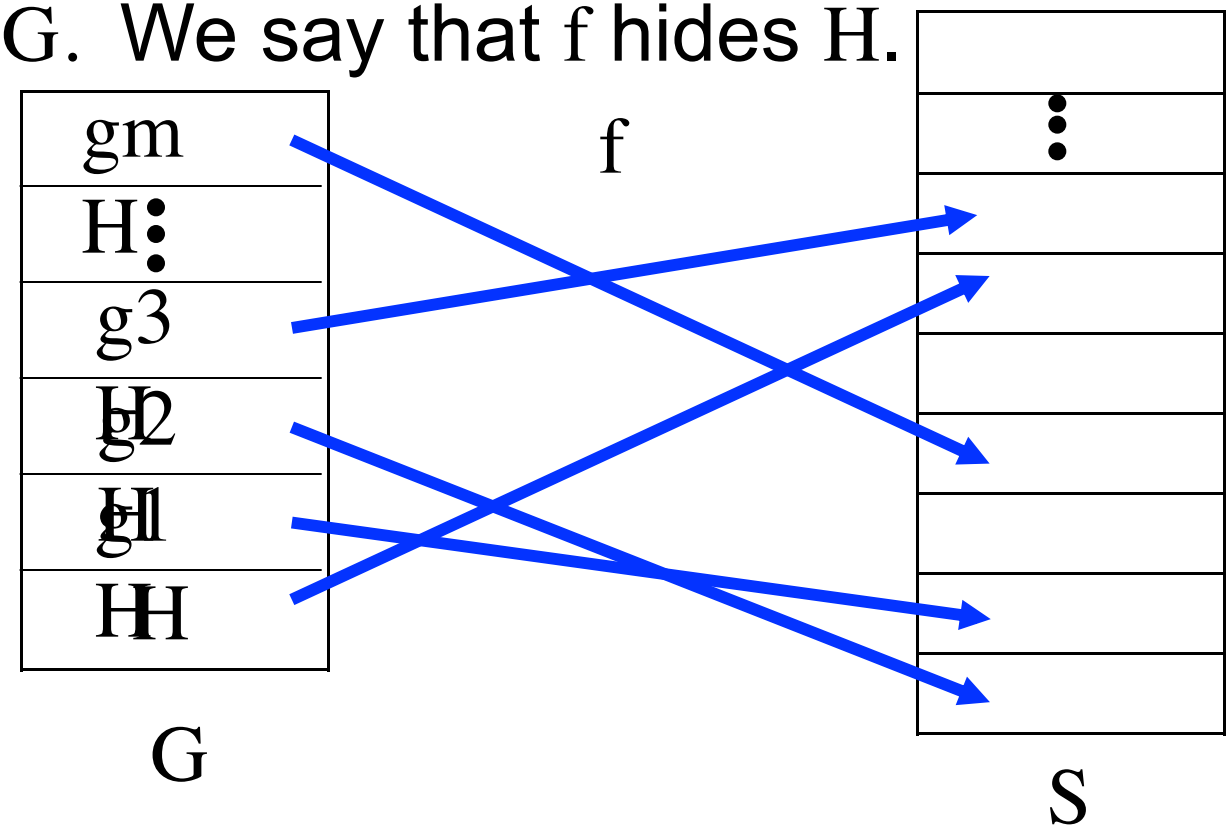
Def: g1H is the set {g1h, h∈H}

g1H is a left coset of G with respect to H.

Given group G, subgroup H, we can uniquely partition G into left cosets of H.

Anatomy of Our Problem

Problem: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that is constant and distinct on left cosets of H in G. We say that f hides H.



The Hidden Subgroup Saga

Why is everyone so interested in the HSP?

. Efficient algorithm for the HSP for Abelian groups

□ ⇒ an efficient algorithm for Factoring (Shor).

2. An efficient algorithm for the HSP for the symmetric group would given an efficient algorithm for graph isomorphism.

3. An efficient algorithm for the HSP for the dihedral group would give algorithms for k-unique shortest vector in a lattice problems (break a cryptosystem)

4. Solving the HSP would make everyone work on really new quantum algorithms.

The Hidden Subgroup Success

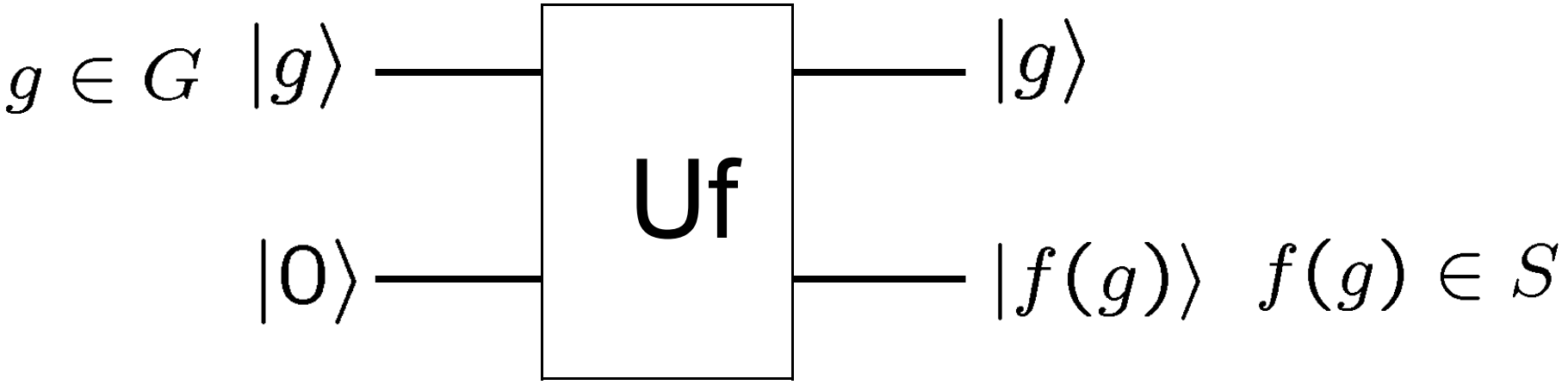
- Abelian groups [Shor 94; Kitaev 95]
- $\mathbb{Z}_2^n \rtimes \mathbb{Z}_2$ [Püschel, Rötteler, Beth 98]
- Normal subgroups [Hallgren, Russell, Ta-Shma 00]
- "Amost Abelian" groups [GSVV 01]
- $\mathbb{Z}_{p^k}^n \rtimes \mathbb{Z}_2$ [FIMMS 02]
- "Near Hamiltonian" groups [Gavinsky 04]
- Some q-hedral groups [MRRS 04]
- Some [Inui, Le Gall 04]
- Heiser $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$ e groups [this talk]

$\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$

Bacon, Childs, and van Dam FOCS 2005

Things are heating up

Standard Query Model



$$\frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|G|}} |g\rangle|f(g)\rangle$$

constant and distinct on different left cosets of H:

$$\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH| \qquad |gH\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |gh\rangle$$

hidden subgroup state

HSP as State Distinction

Single Copy

Given: $\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH|$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*

Multiple Copy

Given: $\rho_H^{\otimes k}$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*

Query Complexity of HSP

Given: *k=O(log(N))* standard queries to HSP

Exists: an inefficient quantum algorithm which

solves any HSP. [Ettinger, Hoyer, Knill 99]

"Query complexity of HSP is polynomial."

Results: Shor's algorithm is optimal with O(1) queries to the HSP [Ip 03]

Results: There are groups for which *O(log(N))* standard queries are necessary [BCV 05]

Holevo's Condition

Given: ρ_i with probability *pi*

Measure: POVM with elements $\{G_j\}$

Outcome: *j* corresponds to guessing state ρ_j

Maximize: probability of successfully identifying ρ_j

Necessary and Sufficient Condition: [Holevo 73]

$$\sum_i p_i \rho_i G_i = \sum_i p_i G_i \rho_i \qquad \sum_i p_i \rho_i G_i \geq p_j \rho_j$$

- **Hard to find explicit forms for *Gj***

Pretty Good

Given: ρ_i with equal probabilities

Observation: the "Pretty Good Measurement" is often pretty good.

Pretty Good Measurement (PGM):

$$G_j = T^{-1/2} \rho_j T^{-1/2} \qquad T = \sum_i \rho_i$$

(plus $I - \sum_j G_j$ if needed.)

Hidden Reflection



Given: $\rho_H^{\otimes k}$ with equal a priori probability

Dihedral HSP

$$H = \{e, rs^l\}, l \in \mathbb{Z}_N$$

Theorem: [Bacon, Childs, van Dam 05]

The PGM is the optimal measurement for identifying *H* (hidden reflection problem)

Theorem: [Moore, Russell 05]

The PGM is the optimal measurement for identifying *H* when *H* and *G* are a Gelfand pair.

arXiv generalization speed = Jan 10 – Jan 31 = 21 days

Must Work Hard



Theorem: [Bacon, Childs, van Dam 05]

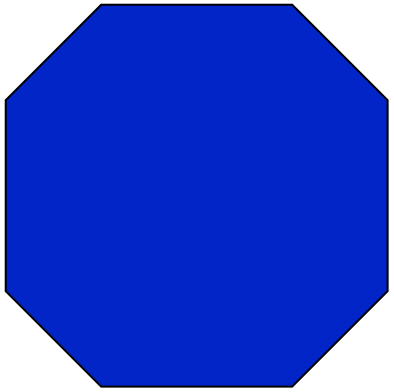
Determining the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

Theorem: [Bacon, Childs, van Dam 05]

Determining the most significant bit of the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

A Strange Lock





Rotation : | 0 | 0 | 0 |

Reflection: | 1 |

Display: | BFD |

Setup: Display will give same result if we enact rotation by k and if we enact rotation by -k+x and a reflection.

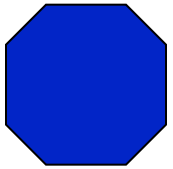
x is the hidden combo.

Goal: Make display repeat to open the lock.

Dihedral Hidden Reflection Problem

Dihedral HSP

Dihedral group



D8

G = DN: symmetry group of a N-sided regular polygon.

Rotation ≡ s, sN=e

Reflection ≡ r, r2=e and rsr=s-1

Element of DN: rtsk, where t∈Z2,s∈ZN

Theorem: Solving the hidden subgroup problem when H={e,rsl}, l∈ZN or H={e} solves the full Dihedral HSP [Etinger, Hoyer 98]

Dihedral HSP Facts

Efficient algorithm for DHSP

[Regev 02]

Efficient algorithm for the poly(d) unique shortest vector in a lattice problem

Quantum algorithm: $2^{O(\sqrt{N})}$ time [Kuperberg 03]
poly(log(N)) space [Regev 04]

Efficient algorithm average case subset sum

[Regev

Efficient quantum Algorithm for DHSP

One Copy

Punishment

One copy of the Hidden Subgroup state:

$$\rho_d = \frac{1}{N} \sum_{k \in \mathbb{Z}_N} |\phi_{d,k}\rangle\langle\phi_{d,k}| \quad H = \{e, rs^d\}, l \in \mathbb{Z}_N$$

$$|\phi_{d,k}\rangle = \frac{1}{\sqrt{2}}(|k, 0\rangle + |-k+d, 1\rangle)$$

Basis char

$$\rho_d = \frac{1}{N} \sum_{k \in \mathbb{Z}_N} |\tilde{\phi}_{d,k}\rangle\langle\tilde{\phi}_{d,k}| \qquad \omega = \exp\left(\frac{2\pi i}{N}\right)$$

$$|\tilde{\phi}_{d,x}\rangle = |x\rangle \frac{1}{\sqrt{2}}\left(|0\rangle + \omega^{xd}|1\rangle\right)$$

$$\rho_d = \frac{1}{2N} \sum_{x \in \mathbb{Z}_N, b, c \in \mathbb{Z}_2} \omega^{(b-c)xd} |x, b\rangle\langle x, c|$$

Multi Copy

More Punishment

$$\rho_d^{\otimes k} = \frac{1}{(2N)^k} \sum_{x \in \mathbb{Z}_N^k} \sum_{b,c \in \mathbb{Z}_2^k} \omega^{((b-c)\cdot x)d} |x, b\rangle\langle x, c|$$

$$\rho_d^{\otimes k} = \frac{1}{(2N)^k} \sum_{x \in \mathbb{Z}_N^k} \sum_{p,q \in \mathbb{Z}_N} \sqrt{\eta_p^x \eta_q^x}\, \omega^{(p-q)d} |S_p^x\rangle\langle S_q^x|$$

$$|S_t^x\rangle := \frac{1}{\sqrt{\eta_t^x}} \sum_{b \in S_t^x} |x, b\rangle \quad S_t^x := \{b \in \mathbb{Z}_2^n | b \cdot x = t\}$$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

Optimal PGM

Stop The Punishment

$$E_j = \frac{1}{N} \sum_{x \in \mathbb{Z}_N^k} \sum_{p,q \in \mathbb{Z}_N} \omega^{(p-q)j} |S_p^x\rangle\langle S_q^x|$$

$$|S_t^x\rangle := \frac{1}{\sqrt{\eta_t^x}} \sum_{b \in S_t^x} |x, b\rangle$$

$$S_t^x := \{b \in \mathbb{Z}_2^n | b \cdot x = t\}$$

Note: Entangled measurement over k copies

Success Probability

Probability of correctly guessing reflection:

$$p = \frac{1}{2^k N^{k+1}} \sum_{x \in \mathbb{Z}_N^k} \left(\sum_{t \in \mathbb{Z}_N} \sqrt{\eta_t^x} \right)^2$$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

p is related to a phase transition

Subset Sum Problem

Given: k integers from $\mathbb{Z}_N := \{0, 1, \ldots, N-1\}$ i.e. $x \in \mathbb{Z}_N^k$, and a target $t \in \mathbb{Z}_N$

Goal: Find a subset of *x* which sums to *t*

$$\sum_{x_i \in S} x_i = t \mod N$$

Problem: does there exists such a subset?

NP-complete

Problem: If such a subset exists, return this subset

NP-hard

basis for old (broken) public key cryptosystem

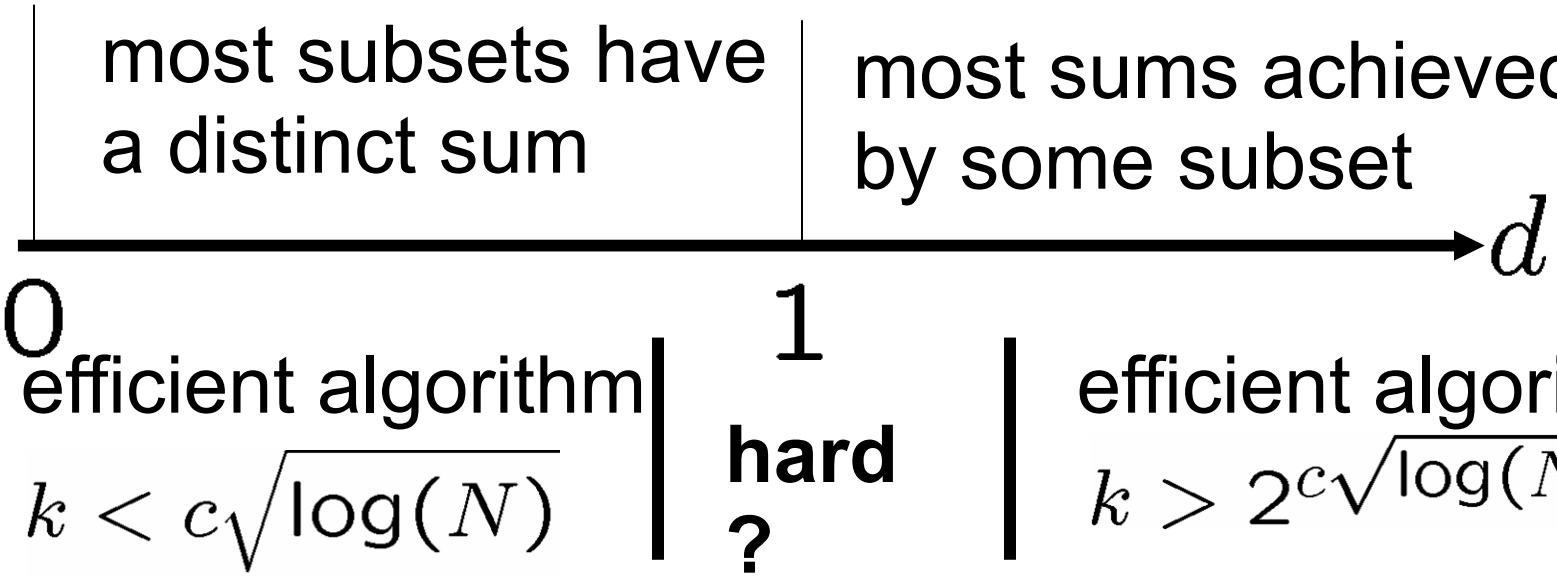
The Subset Sum Landscape

Given: k integers from $\mathbb{Z}_N := \{0, 1, \ldots, N-1\}$

i.e. $x \in \mathbb{Z}_N^k$, and a target $t \in \mathbb{Z}_N$

Define: density := $d = \frac{k}{\log N}$

"low density" **"high density"**



[Lagarias, Odlyzko

[Flaxman, Pryzdatek 04]

Success Probability

Probability of correctly guessing reflection:

$$p = \frac{1}{2^k N^{k+1}} \sum_{x \in \mathbb{Z}_N^k} \left(\sum_{t \in \mathbb{Z}_N} \sqrt{\eta_t^x} \right)^2$$

$\mathbb{Z}_N^k :=$ k dimensional vector of elements of $\mathbb{Z}_N$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

Two orders:

For *k<log N*, *p* is exponentially small (in log(N))

For *k>log N*, *p* is *O(1)*

transition at density = *d=1*

Must Work Hard

Theorem: [Bacon, Childs, van Dam 05]
Determining the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

Theorem: [Bacon, Childs, van Dam 05]
Determining the most significant bit of the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

The Dihedral Hidden Subgroup problem looks hard,
Are there similar groups where the combinatorial problem is not subset sum?

Heisenberg Group

Semidirect product group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Elements of the group:

$$\left(\begin{pmatrix} x \\ y \end{pmatrix}, b \right) \quad x, y, b \in \mathbb{Z}_p$$

Multiplication rule:

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, b_1 \right) \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_2 \right) =$$

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{b_1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_1 + b_2 \right)$$

From the play “Copenhagen” Margrethe Bohr to Heisenberg:

“Your talent is for ski-ing too fast for anyone to see where you are. For always being in more than one position at a time, like one of your particles.”

Heisenberg HSP

We can restrict to subgroups of the form

$$\mathcal{H}_{d_1,d_2} = \left\langle \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right) \right\rangle$$

$$\mathcal{H}_{d_1,d_2} = \left\{ \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right), \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right), \left(\begin{pmatrix} 2d_1 + d_2 \\ 2d_2 \end{pmatrix}, 2 \right), \ldots \right\}$$

General version of theorem from [Etinger, Hoyer 98]

Heisenberg HS State

Restricted subgroups:

$$\mathcal{H}_{d_1,d_2} = \left\{ \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right), \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right), \left(\begin{pmatrix} 2d_1 + d_2 \\ 2d_2 \end{pmatrix}, 2 \right), \ldots \right\}$$

Random coset states:

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| \left(\left(\begin{pmatrix} l_1 \\ l_2 \end{pmatrix} + M^{(b)} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \right), b \right) \right\rangle$$

$$M^{(b)} = \sum_{i=0}^{b-1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^i = \begin{bmatrix} b & \frac{b(b-1)}{2} \\ 0 & b \end{bmatrix}$$

Coset labels: $l_1, l_2 \in_U \mathbb{Z}_p$

Hidden Subgroup states:

$$\rho_{d_1,d_2} = \frac{1}{p^2} \sum_{l_1,l_2 \in \mathbb{Z}_p} |l_1, l_2, d_1, d_2\rangle\langle l_1, l_2, d_1, d_2|$$

Optimal Measumrent

The optimal measurement for k copies is again the pretty good measurement.

For one copy, the probability of successfully identifying the hidden subgroup is

$$Pr(succ) = O\left(\frac{1}{p}\right)$$

But for two copies, the probability of succesfully
identifying the hidden subgroup is

$$Pr(succ) = O(1)$$

Can we efficiently implement this measurement?

HS

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| \left(\left(\begin{pmatrix} l_1 \\ l_2 \end{pmatrix} + M^{(b)} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \right), b \right) \right\rangle$$

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| l_1 + bd_1 + \frac{b(1-b)}{2} d_2, l_2 + bd_2; b \right\rangle$$

Perform the QFT over $(\mathbb{Z}_p)^2$ to the first two registers:

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p^3}} \sum_{b, x_1, y_1 \in \mathbb{Z}_p} \omega_p^{(l_1 + bd_1 + \frac{b(1-b)}{2} d_2) x_1 + (l_2 + bd_2) y_1} |x_1, y_1, b\rangle$$

Measure these two registers:

$$x_1, y_1 \in_U \mathbb{Z}_p$$

$$|x_1, y_1, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \omega_p^{(bd_1 + \frac{b(1-b)}{2} d_2) x_1 + bd_2 y_1} |b\rangle$$

HS

Two copies: $x_1, y_1, x_2, y_2 \in_U \mathbb{Z}_p$

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{(b_1 x_1 + b_2 x_2) d_1 + \left(\frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2 \right) d_2} |b_1, b_2\rangle$$

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{v d_1 + w d_2} |b_1, b_2\rangle$$

Compute v and w:

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{v d_1 + w d_2} |b_1, b_2\rangle |v, w\rangle$$

Uncompute

If we could erase the b1,b2 values:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle \rightarrow \frac{1}{p}|0,0\rangle\sum_{v,w}\omega_p^{vd_1+wd_2}|v,w\rangle$$

Then a QFT over $(\mathbb{Z}_p)^2$ would find d1 and d2.

Remember v and w are a function of b1 and b2.

Solving Equations

Compute v and w:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle$$

$$v = b_1x_1 + b_2x_2$$

$$w = \frac{b_1(1-b_1)}{2}x_1 + b_1y_1 + \frac{b_2(1-b_2)}{2}x_2 + b_2y_2$$

For fixed x1,x2,y1,y2 let $S_{v,w}$ be the set of b1,b2

solutions to these equations

$$\frac{1}{p}\sum_{v,w\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}\left(\sum_{b_1,b_2\in S_{v,w}}|b_1,b_2\rangle\right)|v,w\rangle$$

We want to implement $\frac{1}{\sqrt{|S_{v,w}|}}\sum_{b_1,b_2\in S_{v,w}}|b_1,b_2\rangle|v,w\rangle \rightarrow |0,0\rangle|v,w\rangle$

Solving Equations

$$\frac{1}{\sqrt{|S_{v,w}|}} \sum_{b_1,b_2 \in S_{v,w}} |b_1, b_2\rangle |v, w\rangle \rightarrow |0,0\rangle |v,w\rangle$$

Easier to think about the inverse (do it unitarily):

$$|0,0\rangle |v,w\rangle \rightarrow \frac{1}{\sqrt{|S_{v,w}|}} \sum_{b_1,b_2 \in S_{v,w}} |b_1, b_2\rangle |v, w\rangle$$

i.e. given v and w, transform to superposition over solutions to the equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Solving Equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Assuming y1,y2,y1+y2 do not equal 0 (prob O(1/p)),
then

$\frac{p(p-1)}{2}$ solutions have $|S_{v,w}| = 0$

p solutions have $|S_{v,w}| = 1$

$\frac{p(p-1)}{2}$ solutions have $|S_{v,w}| = 2$

Solving Equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Focus on when there are two solutions.

We can use standard finite arithmetic to find these solutions. Use to create labeled superposition of solutions:

$$\frac{1}{\sqrt{2}} \sum_{j=0,1} |j, (b_1)_j, (b_2)_j\rangle$$

Hadamard first qubit, measure. If 0, you have

$$\frac{1}{\sqrt{2}} \sum_{j=0,1} |(b_1)_j, (b_2)_j\rangle$$

Heisenberg HSP Solved

This allows us to perform:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle \rightarrow \frac{1}{p}|0,0\rangle\sqrt{|S_{v,w}|}\sum_{v,w}\omega_p^{vd_1+wd_2}|v,w\rangle$$

Which has a O(1/2) fidelity squared with the state we desire.

QFTing the second register will let us find d1 and d2

Efficient algorithm for the Heisenberg HSP!

Extensions

This algorithm can be extended to an efficient quantum algorithm for all $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$

Also, certain metacyclic groups $\mathbb{Z}_N \rtimes \mathbb{Z}_p$

Interesting

Entangled measurement over two registers used
in a constructive form.

Superposition over solutions to equations we can efficiently invert.

Open

The PGM is not optimal for all groups, what is the
optimal multicopy measurement?

Other groups for which this technique works?

Connections to the Clebsch-Gordan transform over finite groups. When can these transforms be efficiently implemented?

Conclusion

Subgroups fo s are necessary.

PGM is often g HS states.

Efficient algo generalizatio and metacycl

